

Benchmarking Thermodynamic Block Gibbs Sampling against Metropolis-Hastings and Single-Site Gibbs for Ground State Estimation in Classical Optimisation Problems

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Abstract—Extropic’s thermodynamic computers harness thermal noise as a computational resource, with rejection-free block Gibbs sampling as its algorithmic core. Their THRML library simulates this approach on conventional GPUs but targets AI workloads such as energy-based and diffusion-like models. This thesis evaluates whether THRML’s block Gibbs transfers to classical combinatorial optimisation, benchmarking it on a GPU against Metropolis-Hastings (MH) and single-site Gibbs across ferromagnetic Ising, anti-ferromagnetic Ising, and MaxCut instances on 2D lattices (up to 100×100 nodes) and random regular graphs (up to 2,500 nodes), with a sweep over inverse temperature β . Block Gibbs achieves two to three orders of magnitude higher effective samples per second (ESS/s) than both baselines while matching or exceeding MH in solution quality on most instances. The advantage is strongest on dense MaxCut (degree-5), where effect sizes grow monotonically with problem size (rank-biserial r up to 0.34); the lone exception is the strongly ordered regime ($\beta = 2.0$), where MH’s rejection step yields a small but significant edge. These results support the algorithmic premise of thermodynamic sampling and identify dense MaxCut as the problem class where block Gibbs benefits most.

Index Terms—thermodynamic computing, block Gibbs sampling, Metropolis-Hastings, Markov chain Monte Carlo, Ising model, MaxCut, combinatorial optimisation, ground state estimation, THRML, probabilistic computing, energy-based models

I. INTRODUCTION

Solving hard combinatorial optimisation problems is central to applications spanning logistics, drug discovery, and machine learning, yet the energy cost of running these workloads on conventional hardware continues to grow. Thermodynamic computing has emerged as a promising alternative paradigm: rather than suppressing thermal noise, it harnesses it as a computational resource [7], [8]. Here we study this paradigm at the software level, benchmarking GPU implementations of block and single-site MCMC samplers on canonical discrete optimisation problems.

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The Ising model is the canonical test bed for combinatorial solvers. It assigns a binary spin $s_i \in \{-1, +1\}$ to each node of a graph, with an energy function whose minimisation is NP-hard in general. Any NP problem can be mapped to an Ising instance with only polynomial overhead [1], making progress on Ising solvers broadly significant. This thesis studies two regimes: the *ferromagnetic* case ($J_{ij} > 0$), where aligned spins are favoured and the energy landscape is relatively smooth, and the *anti-ferromagnetic* case ($J_{ij} < 0$), which produces frustrated, rugged landscapes. The MaxCut problem—partition a graph to maximise the number of crossing edges—maps exactly to anti-ferromagnetic Ising and serves as a natural stress test for any sampler [11].

The standard baseline for ground state estimation is Metropolis-Hastings (MH), whose rejection step allows the Markov chain to escape local minima and guarantees convergence to the target distribution. Gibbs sampling eliminates rejection by sampling each spin directly from its conditional distribution, removing the accept/reject overhead but preventing the chain from crossing low-probability barriers—a failure mode that is particularly severe in frustrated anti-ferromagnetic systems [4]. Standard single-site Gibbs serves as a natural intermediate baseline: it shares the rejection-free property of block Gibbs but without parallelism, isolating the contribution of block updates. Block Gibbs (THRML) extends this by using graph colouring to identify conditionally independent spins that can be updated simultaneously as a block, gaining throughput over single-site Gibbs while retaining this fundamental limitation.

Extropic’s Thermodynamic Sampling Unit (TSU) is built around block Gibbs, deliberately forgoing rejection in exchange for the energy efficiency that probabilistic hardware enables. Jelinčič et al. project performance parity with GPUs at roughly 10,000 times lower energy [7], building on earlier work on p-bit circuits [5] and sparse Ising Machines [6]. The THRML library [9] simulates TSU-style block Gibbs on a conventional GPU, primarily targeting AI workloads such as energy-based models and diffusion-like generative

models. This thesis applies THRML to classical combinatorial optimisation—Ising ground-state estimation and MaxCut—to evaluate how well its block Gibbs approach transfers beyond its intended AI domain.

The broader landscape of Ising solvers spans a wide range of hardware paradigms [1]. Quantum annealers (e.g. D-Wave) and optical Coherent Ising Machines reach high speeds but require specialised hardware, and annealers are further restricted to sparse, hardware-native topologies. On the probabilistic hardware side, p-bit circuits [5] and sparse Ising Machines [6] implement rejection-free Boltzmann sampling with stochastic binary units, sharing the core philosophy of block Gibbs. Among purely software approaches, GPU-accelerated MCMC [2], [3] represents the accessible state-of-the-art: it exploits the local neighbourhood structure of spin models to parallelise proposals across thousands of CUDA cores. THRML bridges these paradigms by simulating probabilistic hardware-style block Gibbs on a conventional GPU, making it directly comparable to GPU MCMC baselines without requiring specialised hardware.

Despite these advances, the empirical performance trade-offs of block Gibbs across different problem classes remain incompletely characterised across practical problem scales. It is unclear whether THRML block Gibbs achieves solution quality comparable to MH, and under which conditions its computational efficiency advantage holds. This thesis addresses that gap through a systematic three-way comparison of THRML block Gibbs, standard single-site Gibbs, and Metropolis-Hastings, all running on identical GPU hardware. Experiments cover two graph topologies—2D square lattice and random regular graph (RRG)—across ferromagnetic Ising, anti-ferromagnetic Ising, and MaxCut instances. Grid sizes range from 50×50 to 100×100 ; RRG sizes range from 900 to 2,500 nodes. MaxCut is evaluated at graph degrees 3 and 5 to test the effect of connectivity. An additional sweep over inverse temperature $\beta \in \{0.1, 0.5, 1.0, 2.0\}$ on the 70×70 grid probes sampler behaviour at low and high temperatures.

Three research questions guide the work:

- 1) How do the three samplers compare in solution quality across problem types and system sizes?
- 2) How does their relative performance scale with graph topology and temperature?
- 3) What are the computational efficiency trade-offs, as measured by effective samples per second and Gelman–Rubin convergence diagnostics?

The remainder of this thesis is structured as follows: Section II describes the three sampler implementations; Section III defines the experimental methodology; Section IV presents results; Section V interprets findings in the context of existing literature; and Section VI draws conclusions.

II. METHODS

A. Problem Formulation

All three samplers operate on the same shared graph representation. An Ising model instance is defined by a graph

$G = (V, E)$ with $n = |V|$ nodes, edge-wise coupling strengths $J_{ij} \in \mathbb{R}$, and inverse temperature $\beta > 0$. Each node i carries a spin $s_i \in \{-1, +1\}$, and the energy of a configuration $\mathbf{s}$ is:

$$\mathcal{E}(\mathbf{s}) = -\beta \sum_{(i,j) \in E} J_{ij} s_i s_j \quad (1)$$

The samplers target the Boltzmann distribution $P(\mathbf{s}) \propto \exp(-\mathcal{E}(\mathbf{s}))$. For ferromagnetic instances $J_{ij} = +1$, aligned spins are energetically favoured; for anti-ferromagnetic instances $J_{ij} = -1$, opposing spins are favoured. MaxCut is handled through the same formulation by setting $J_{ij} = -1$, so that minimising (1) is equivalent to maximising the number of cut edges. The cut value is computed separately as $\text{Cut}(\mathbf{s}) = \sum_{(i,j) \in E} |J_{ij}| (1 - s_i s_j)/2$, which for unit weights ($|J_{ij}| = 1$) counts the number of cut edges.

B. Metropolis-Hastings

At each step, a node i is chosen uniformly at random and a spin flip is proposed. The energy change from flipping s_i depends only on the local neighbourhood $\mathcal{N}(i)$:

$$\Delta \mathcal{E} = 2\beta s_i \sum_{j \in \mathcal{N}(i)} J_{ij} s_j \quad (2)$$

The flip is accepted with probability $\min(1, \exp(-\Delta \mathcal{E}))$. One sweep consists of n such steps. The acceptance step guarantees detailed balance and allows the chain to cross low-probability barriers, at the cost of rejected proposals that leave the state unchanged.

The implementation uses a padded neighbour list of shape $[n, d_{\max}]$, where $d_{\max}$ is the maximum node degree, with a boolean mask to handle varying degree. The inner loop over n steps is compiled with `jax.lax.scan`. The sampler is JIT-compiled with `jax.jit` and parallelised over independent chains with `jax.vmap` [10].

C. Standard Gibbs Sampling

Gibbs sampling draws each spin exactly from its conditional distribution given all neighbours, eliminating the accept/reject step entirely. For node i :

$$P(s_i = +1 \mid \mathbf{s}_{\setminus i}) = \sigma \left(2\beta \sum_{j \in \mathcal{N}(i)} J_{ij} s_j \right) \quad (3)$$

where σ is the sigmoid function. One sweep consists of n single-node conditional updates. The implementation shares the same neighbour list structure and `lax.scan/vmap` pattern as MH; the only difference is replacing the Metropolis accept/reject with direct conditional sampling from (3).

D. Block Gibbs via THRML

Block Gibbs extends single-site Gibbs by updating entire groups of conditionally independent nodes in parallel. Two nodes are conditionally independent given all others if and only if they share no edge, so the natural grouping is a proper

graph colouring: nodes of the same colour can be sampled simultaneously without violating the Markov property.

The graph is coloured using the DSATUR greedy algorithm [13], producing colour classes $C_1, \dots, C_k$. One block Gibbs step cycles through all classes in order; within each class, all spins are sampled in parallel from their conditionals (3). A 2D square grid is bipartite and requires two colours; a degree-3 random regular graph typically requires three. More colours mean smaller parallel blocks but the same total work per step.

This sampler is implemented using the THRML library [9], which provides an `IsingEBM` energy model, an `IsingSamplingProgram` encoding the colour-block structure, and a `sample_states` function that executes the schedule. The same `jax.vmap` parallelism over independent chains is applied externally.

E. Shared Infrastructure and Timing

All three samplers share a common warm-up phase of n_{warmup} sweeps that are discarded before sample collection. Samples are then collected every n_{thinning} sweeps to reduce autocorrelation. Wall-clock time is measured using `jax.block_until_ready` to ensure all GPU computation is complete before the timer stops, and a separate untimed compilation run precedes each timed run to exclude JIT overhead. This ensures wall-clock times reflect steady-state throughput and are directly comparable across samplers.

III. EXPERIMENTS

The experiments are designed to answer the three research questions by varying problem type, graph topology, system size, and temperature independently, while holding all sampler hyperparameters fixed across conditions. This section describes the experimental design, the configurations tested, and the metrics collected.

A. Experimental Design

Each experiment runs all three samplers—MH, standard Gibbs, and THRML block Gibbs—on the same problem instance with equal chain count and sample budget, and independently sampled random initialisations. This ensures differences in outcome reflect algorithmic behaviour rather than initialisation or resource differences.

All experiments were run on a single laptop NVIDIA RTX 3060 (6 GB VRAM). Each sampler was given $n_{\text{chains}} = 50$ independent parallel chains, initialised from random spin configurations. Each chain ran $n_{\text{warmup}} = 200$ warm-up sweeps (discarded), then collected $n_{\text{samples}} = 500$ samples with $n_{\text{thinning}} = 5$ sweeps between each sample. Each configuration was repeated for $N_{\text{runs}} = 3$ independent runs; chains are pooled across runs (150 total per sampler) for statistical analysis.

B. Problem Instances and Configurations

Four groups of configurations were benchmarked, covering the three problem types and three graph topologies.

1) *Ferromagnetic and Anti-ferromagnetic Grid*: A 2D square lattice with $J = +1$ (ferromagnetic) and $J = -1$ (anti-ferromagnetic) at $\beta = 1.0$. System sizes of 50×50 , 70×70 , and 100×100 are tested to measure scaling behaviour; these sizes were chosen to span a range where critical slowing down becomes progressively more severe. The 2D grid is bipartite, so THRML requires only two colour classes regardless of size.

2) *Temperature Sweep*: A ferromagnetic grid ($J = +1$) at inverse temperatures $\beta \in \{0.1, 0.5, 1.0, 2.0\}$ on a fixed 70×70 lattice, sweeping from the high-temperature disordered phase ($\beta = 0.1$) to the strongly ordered phase ($\beta = 2.0$). The 70×70 size was chosen as a middle ground large enough to make the ordered phase hard to mix on, revealing whether THRML's throughput advantage holds across the full temperature range rather than at a single operating point.

3) *Random Regular Graph*: A random regular graph of degree $d = 3$ with ferromagnetic coupling $J = +1$ at $\beta = 1.0$. Node counts of 900, 1,600, and 2,500 are tested to measure scaling behaviour; these sizes were chosen to span a comparable node-count range to the grid experiments. Unlike the grid, a degree-3 RRG is not bipartite and requires three colour classes under DSATUR (rising to four at degree 5).

4) *MaxCut*: MaxCut on a random regular graph ($J = -1$) at $\beta = 1.0$, evaluated at degrees $d \in \{3, 5\}$ and node counts of 900, 1,600, and 2,500. The degree sweep isolates the effect of graph connectivity on block Gibbs parallelism: higher degree increases the chromatic number, reducing colour-class size and therefore the parallelism available to THRML.

C. Metrics

Four metrics are recorded for each configuration and sampler:

- **Solution quality**: the mean best energy (or cut value for MaxCut) found across chains, and the distribution of per-chain best values. Statistical significance of differences between samplers is assessed using the Mann-Whitney U test on the 150 pooled per-chain best values (3 runs $\times$ 50 chains), with effect size reported as the rank-biserial correlation r .
- **Convergence**: the Gelman-Rubin $\hat{R}$ statistic [12] computed on the energy trajectory (cut value for MaxCut) across all 150 chains. Values $\hat{R} < 1.1$ indicate convergence; values substantially above 1.1 indicate chains have not mixed.
- **Sampling efficiency**: the effective sample size (ESS) of the per-chain energy series (cut value for MaxCut), estimated via integrated autocorrelation time using the FFT [14], and ESS per second as the primary throughput metric.
- **Wall-clock time**: mean wall time across the three runs, excluding JIT compilation.

IV. RESULTS

Results are reported across four experiment groups: grid ferromagnetic, grid anti-ferromagnetic, ferromagnetic random regular graph (RRG), and MaxCut on RRG at degrees 3 and

5. For each group, solution quality (mean best energy or cut value), convergence diagnostics ($\hat{R}$), and sampling efficiency (ESS/s) are reported. Statistical comparisons use the Mann-Whitney U test on 150 pooled per-chain best values (3 runs $\times$ 50 chains); effect sizes are reported as rank-biserial correlation r .

TABLE I
STATISTICAL COMPARISON OF THRML AGAINST MH AND STANDARD GIBBS. SIGNIFICANCE: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, NS = NOT SIGNIFICANT. r = RANK-BISERIAL EFFECT SIZE. $\uparrow$ = THRML ACHIEVES BETTER OUTCOME (LOWER ENERGY FOR ISING, HIGHER CUT FOR MAXCUT); $\downarrow$ = THE OTHER SAMPLER ACHIEVES BETTER OUTCOME. DIRECTION SHOWN ONLY WHEN SIGNIFICANT ($p < 0.05$).

Config	THRML vs MH		THRML vs Gibbs	
	sig	r	sig	r
ferro 50 $\times$ 50	ns	—	ns	—
ferro 70 $\times$ 70	ns	—	*** $\uparrow$	-0.31
ferro 100 $\times$ 100	ns	—	*** $\uparrow$	-0.48
antiferro 50 $\times$ 50	ns	—	* $\uparrow$	-0.11
antiferro 70 $\times$ 70	ns	—	*** $\uparrow$	-0.22
antiferro 100 $\times$ 100	** $\uparrow$	-0.21	*** $\uparrow$	-0.52
$\beta = 0.1$	ns	—	ns	—
$\beta = 0.5$	ns	—	*** $\uparrow$	-0.31
$\beta = 1.0$	ns	—	*** $\uparrow$	-0.25
$\beta = 2.0$	* $\downarrow$	0.13	** $\uparrow$	-0.19
RRG 900	ns	—	ns	—
RRG 1600	* $\uparrow$	-0.04	** $\uparrow$	-0.07
RRG 2500	ns	—	ns	—
MaxCut d3 900	* $\uparrow$	0.15	* $\uparrow$	0.16
MaxCut d3 1600	ns	—	ns	—
MaxCut d3 2500	** $\uparrow$	0.20	*** $\uparrow$	0.22
MaxCut d5 900	* $\uparrow$	0.17	ns	—
MaxCut d5 1600	* $\uparrow$	0.17	* $\uparrow$	0.14
MaxCut d5 2500	*** $\uparrow$	0.27	*** $\uparrow$	0.34

A. Grid: Ferromagnetic and Anti-ferromagnetic

Table I summarises the statistical comparisons. On ferromagnetic grids, solution quality differences between THRML and MH are not significant at any size ($p > 0.25$). Both THRML and MH significantly outperform standard Gibbs at 70 $\times$ 70 and 100 $\times$ 100 ($p < 0.0001$, $|r| = 0.30$ – 0.48), with the gap widening with system size. On anti-ferromagnetic grids the same pattern holds, with the additional result that at 100 $\times$ 100 THRML significantly outperforms MH ($p = 0.0012$, $r = -0.214$).

Gelman-Rubin diagnostics reveal that all three samplers fail to converge within the sample budget on grid instances: $\hat{R}$ ranges from 1.18 to 1.82 across all grid configurations (Figure 1). Figure 2 shows the energy trace for the 100 $\times$ 100 ferromagnetic grid: all three samplers are still descending at the end of the sample window, confirming incomplete mixing. Despite this shared failure to converge, sampling efficiency differs by orders of magnitude: at 50 $\times$ 50, THRML achieves 604.7 ESS/s compared to 0.8 for MH and 0.1 for standard Gibbs—a 756 $\times$ gap over MH. At 100 $\times$ 100, THRML achieves 13.8 ESS/s while both MH and Gibbs fall below 0.05 ESS/s. Wall-clock time scales steeply for MH and Gibbs as system size grows (Figure 12), while THRML’s block parallelism partially offsets this cost.

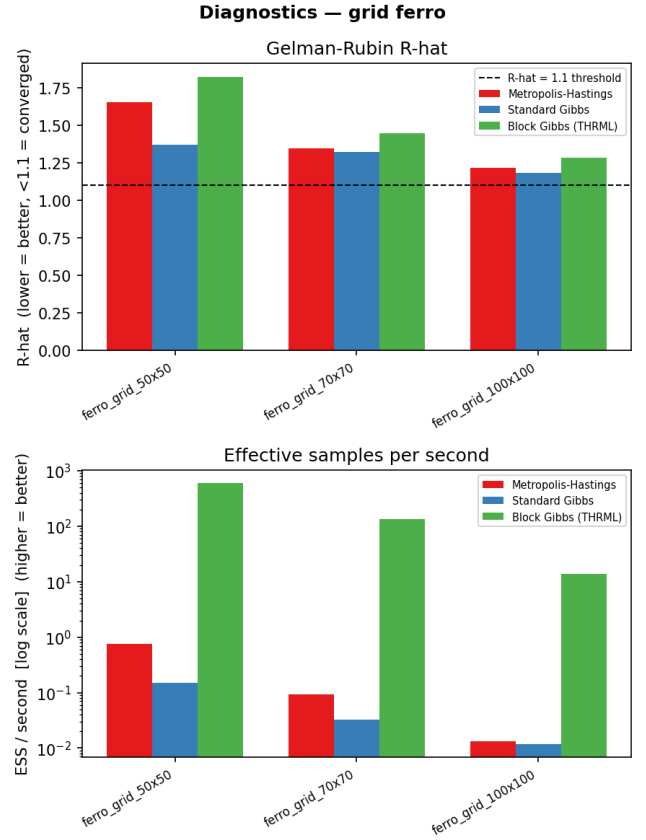


Fig. 1. Gelman-Rubin $\hat{R}$ (top) and ESS/s on log scale (bottom) for the ferromagnetic grid across system sizes. Dashed line indicates the $\hat{R} = 1.1$ convergence threshold.

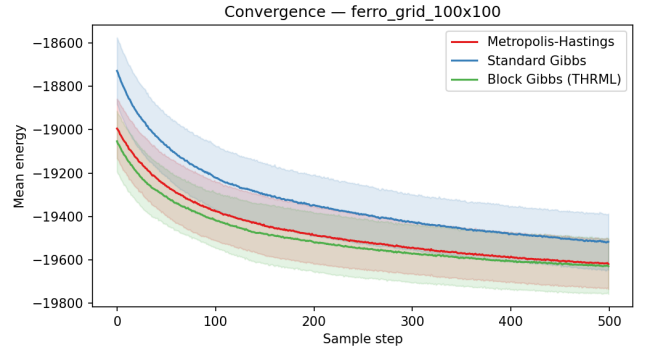


Fig. 2. Mean energy trace (± 1 std across chains) for the 100 $\times$ 100 ferromagnetic grid. All three samplers continue to descend throughout the sample window, indicating incomplete mixing within the sample budget.

B. Temperature Sweep

At high temperature ($\beta = 0.1$), all three samplers converge fully ($\hat{R} = 1.000$) and produce statistically equivalent solution quality ($p > 0.06$). THRML achieves 3,462 ESS/s at $\beta = 0.1$ compared to 2.9 for MH and 2.8 for standard Gibbs—a 1,194 $\times$ gap. As temperature decreases, $\hat{R}$ rises for all samplers and ESS/s falls. At $\beta = 2.0$, MH significantly outperforms THRML in solution quality ($p = 0.028$, $r = 0.125$), while

both outperform standard Gibbs ($p < 0.003$). Figure 3 shows solution quality and wall-clock time as functions of β .

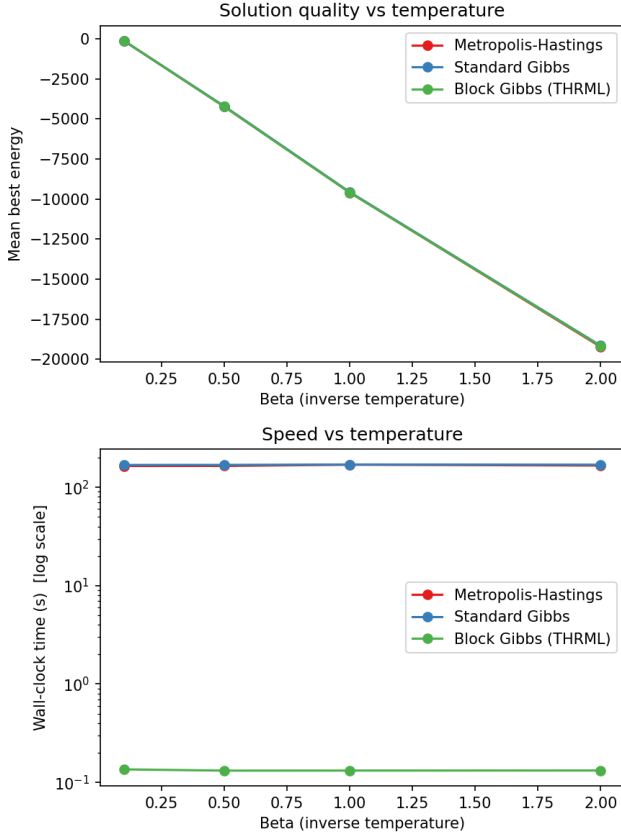


Fig. 3. Mean best energy (top) and wall-clock time on log scale (bottom) vs. inverse temperature β on a 70×70 ferromagnetic grid.

C. Random Regular Graph

All three samplers converge fully on ferromagnetic RRG instances: $\hat{R} \leq 1.001$ across all sizes, providing a clean baseline for comparing throughput. At 900 nodes all samplers find identical mean best energy (-1350.00 , $p = 1.0$); at 1,600 nodes the difference reaches significance ($p = 0.032$ vs. MH, $p = 0.003$ vs. Gibbs) but the effect is negligible ($r = -0.040$ and $r = -0.067$); and at 2,500 nodes differences are not significant. In practical terms, solution quality is therefore equivalent across all three samplers at every size. By contrast, sampling efficiency differs sharply: THRML ESS/s is $325\times$ higher than MH at 900 nodes (4,527 vs. 13.9) and $781\times$ higher at 2,500 nodes (3,904 vs. 5.0). Figure 4 shows how ESS/s scales with system size for each sampler.

D. MaxCut

1) *Degree 3*: $\hat{R}$ values remain below 1.04 for all samplers at all sizes, indicating near-convergence. THRML achieves higher mean best cut value at 900 nodes ($p = 0.027$, $r = 0.147$) and 2,500 nodes ($p = 0.003$, $r = 0.199$; vs. Gibbs $p = 0.0009$, $r = 0.222$); differences are not significant at 1,600 nodes. MH and standard Gibbs are statistically

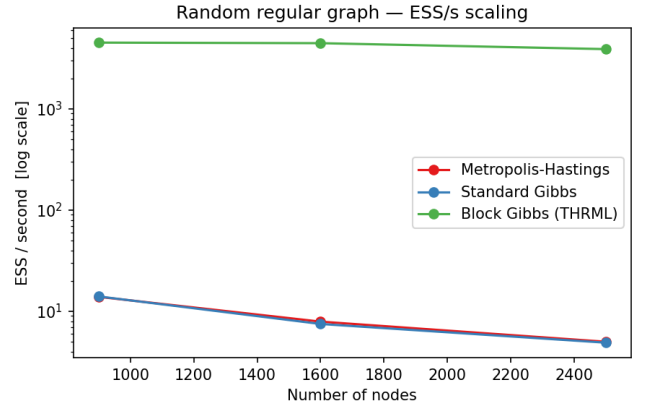


Fig. 4. ESS/s vs. number of nodes on log scale for the ferromagnetic RRG. THRML maintains two to three orders of magnitude higher throughput than MH and standard Gibbs at all sizes, while solution quality is near-identical.

equivalent at all sizes. THRML ESS/s exceeds that of MH by factors of 470 to 1,060 across the three sizes.

2) *Degree 5*: MaxCut degree-5 produces the strongest and most consistent quality advantage for THRML (Figure 5). THRML significantly outperforms MH at all three sizes ($p = 0.013$, $r = 0.165$ at 900 nodes; $p = 0.012$, $r = 0.168$ at 1,600 nodes; $p = 0.0001$, $r = 0.265$ at 2,500 nodes). THRML also outperforms standard Gibbs at 1,600 and 2,500 nodes ($p = 0.038$, $r = 0.138$; $p < 0.0001$, $r = 0.336$). Effect sizes grow monotonically with problem size for both comparisons: vs. MH, r increases from 0.165 to 0.168 to 0.265; vs. Gibbs, from 0.094 to 0.138 to 0.336. $\hat{R}$ remains below 1.09 for all samplers. Figure 6 shows ESS/s: THRML achieves 671, 386, and 229 ESS/s at sizes 900, 1,600, and 2,500 respectively, compared to below 2.0 for MH and Gibbs at all sizes—a gap of $350\times$ to $760\times$.

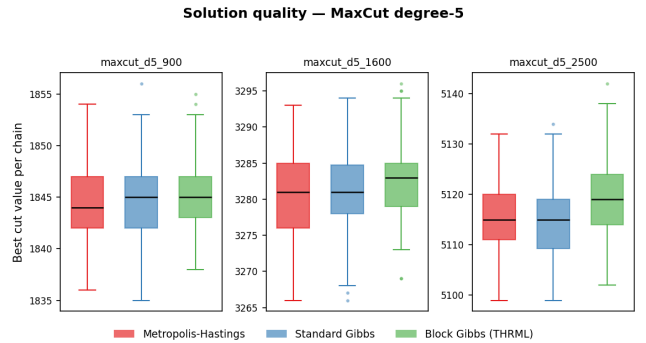


Fig. 5. Distribution of per-chain best cut values across samplers for MaxCut degree-5, one panel per system size with independent vertical axes. Box = interquartile range, central line = median, whiskers = $1.5 \times \text{IQR}$, points = outliers. Higher is better; THRML's median rises above both baselines, most clearly at 2,500 nodes.

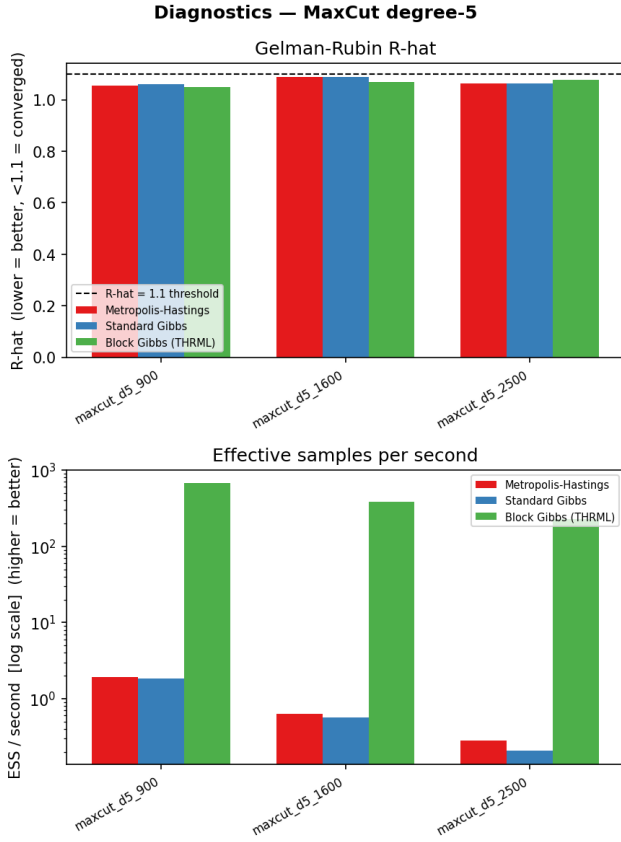


Fig. 6. Gelman-Rubin $\hat{R}$ (top) and ESS/s on log scale (bottom) for MaxCut degree-5 across system sizes.

V. DISCUSSION

A. Sampling Efficiency

The most consistent finding across all experiment groups is that THRML achieves dramatically higher ESS/s than both MH and standard Gibbs—typically two to three orders of magnitude. This advantage is driven primarily by wall-clock throughput rather than by faster decorrelation. The per-chain effective sample size is comparable between THRML and MH (within a factor of 1.0–1.6 across all configurations; Table II), so the orders-of-magnitude gap in ESS/s reflects how much faster THRML produces those samples: block Gibbs updates an entire colour class in parallel on the GPU, costing a handful of vectorised operations per sweep, whereas MH and single-site Gibbs perform n sequential single-spin updates per sweep. Against single-site Gibbs, THRML also enjoys a secondary decorrelation benefit (ESS up to $5\times$ higher on small grids), consistent with the theoretical motivation that simultaneous colour-class updates break the local correlations that slow single-site mixing [4]; against MH, however, the gain is almost entirely one of throughput.

B. Solution Quality

THRML matches MH in solution quality across the majority of configurations and outperforms standard Gibbs on all large

grid instances. This is a strong result: THRML forgoes the rejection step entirely, yet this does not lead to systematically worse solutions. The exception at $\beta = 2.0$ —where MH significantly outperforms THRML ($p = 0.028$, $r = 0.125$)—is consistent with the known limitation of Gibbs-based samplers at very low temperatures. At high β the energy landscape is sharply peaked, and MH’s rejection step allows it to reject energy-raising moves, whereas the rejection-free Gibbs samplers always accept and can settle into slightly suboptimal configurations. This is the same barrier-crossing mechanism identified by Roussel et al. in Restricted Boltzmann Machines [4], here observed on the Ising model.

Standard Gibbs consistently underperforms on large grid instances ($|r| = 0.22$ – 0.52 vs. THRML). Single-site random-order updates on a lattice produce strongly correlated samples because adjacent spins are updated sequentially rather than in parallel, preserving local correlations across steps. Block Gibbs eliminates this by updating all spins of a colour class simultaneously, which is why the quality gap between THRML and standard Gibbs grows with system size.

C. Convergence

The Gelman-Rubin diagnostics reveal an important limitation: all three samplers fail to satisfy $\hat{R} < 1.1$ on grid instances within the sample budget of 500 samples. This is not unique to any one sampler—it reflects the intrinsic difficulty of mixing on large 2D lattices, where the correlation length grows with system size. Despite THRML’s superior ESS/s, the absolute number of effective samples at 100×100 ($\text{ESS} \approx 4.8$ per chain) remains low in absolute terms. Longer runs or larger sample budgets would be needed to achieve convergence on the largest grid instances.

In contrast, all three samplers converge fully on RRG instances ($\hat{R} \leq 1.001$), and MaxCut instances show near-convergence ($\hat{R} < 1.09$). The difference is structural: RRG and MaxCut graphs lack the large, slowly-rearranging aligned domains that form on the ordered 2D lattice.

D. Effect of Graph Connectivity

The MaxCut degree sweep reveals that THRML’s quality advantage grows with graph degree: effect sizes are consistently larger at degree 5 than degree 3, and grow monotonically with problem size at degree 5. This is counterintuitive at first as higher degree increases the chromatic number, reducing the size of each parallel colour block and therefore THRML’s parallelism. However, denser graphs create more frustrated interactions, making the energy landscape more rugged and the sampling problem harder for all methods. In this regime, THRML’s faster decorrelation per step translates into a larger practical quality advantage, suggesting block Gibbs is particularly well suited to hard frustrated problems.

E. Implications for Thermodynamic Hardware

Our results provide a systematic empirical characterisation of the algorithmic behaviour of TSU-style block Gibbs across a range of problem classes. The finding that block Gibbs

achieves equivalent or superior solution quality to MH at dramatically higher ESS/s supports the algorithmic premise of the TSU design. Importantly, these results represent a conservative lower bound on the benefit of thermodynamic sampling hardware: THRML runs on a conventional GPU, which is not architecturally optimised for probabilistic computation. On dedicated TSU silicon, the same algorithmic efficiency advantage would compound with hardware-level energy savings, making the practical case stronger than the GPU simulation alone suggests. On the basis of these results, the TSU’s rejection-free block Gibbs approach appears well suited to MaxCut and ferromagnetic optimisation workloads. The $\beta = 2.0$ anomaly suggests caution for strongly ordered low-temperature regimes, where rejection-based methods retain a quality advantage.

F. Limitations

All experiments were conducted on a single laptop NVIDIA RTX 3060 (6 GB VRAM). Results may differ on hardware with different memory bandwidth or parallelism characteristics. The sample budget (500 samples, 200 warm-up sweeps) is sufficient for RRG and MaxCut but insufficient for full convergence on large grid instances. Energy consumption was not directly measured; ESS/s is used as a proxy for energy efficiency, following the assumption that throughput advantage translates to energy advantage at equivalent solution quality [7]. The MH and single-site Gibbs baselines update one spin at a time within each chain and are not parallelised across spins as in checkerboard GPU MH [3]; the ESS/s comparison therefore quantifies the benefit of block parallelism over sequential single-site sampling, and a spin-parallel MH would reduce the wall-clock gap. Each configuration also uses a single graph instance (fixed seed) rather than an average over multiple random graphs of the same size and degree, and ESS is estimated from the energy (or cut) trajectory only. The three samplers were compared under an unpaired design—each draws its own random initial states rather than sharing them—so a paired design with common initialisations and a Wilcoxon signed-rank test would have removed chain-to-chain initialisation variance and afforded greater statistical power. Finally, all results reflect THRML running as a software simulation on GPU rather than on dedicated thermodynamic hardware; the efficiency advantages observed here do not include the projected hardware-level energy savings of the TSU.

VI. CONCLUSION

This thesis presented a systematic three-way comparison of THRML block Gibbs, standard single-site Gibbs, and Metropolis-Hastings on ferromagnetic Ising, anti-ferromagnetic Ising, and MaxCut problems across two graph topologies and a range of system sizes and temperatures.

RQ1 — Solution quality. THRML achieves solution quality statistically equivalent to MH across the majority of configurations, and significantly outperforms standard Gibbs on all large grid instances. On MaxCut degree-5, THRML

significantly outperforms MH at all three system sizes and standard Gibbs at the two larger sizes, with effect sizes growing monotonically with problem size. The single exception is the very low temperature regime ($\beta = 2.0$), where MH’s rejection step confers a small but significant quality advantage. Overall, forgoing rejection does not lead to systematically worse solutions.

RQ2 — Scaling with topology and temperature. At high temperature ($\beta = 0.1$) all three samplers are statistically indistinguishable in quality; differences emerge as temperature decreases and the energy landscape becomes more rugged. On 2D grids, none of the samplers fully converge within the sample budget, reflecting the intrinsic difficulty of mixing on lattices. On random regular graphs all three converge fully, and on MaxCut they reach near-convergence; THRML’s quality advantage grows with graph degree, suggesting block Gibbs is particularly well suited to dense frustrated problems.

RQ3 — Computational efficiency. THRML achieves two to three orders of magnitude higher ESS/s than MH and standard Gibbs across all configurations. On the fully-converged instances—where both ESS and wall-clock time are reliably estimated—the gap ranges from $325\times$ over MH on the smallest RRG to $1,194\times$ in the high-temperature ($\beta = 0.1$) regime. This efficiency advantage holds even in cases where solution quality is equivalent, meaning THRML reaches the same quality with far fewer wall-clock seconds.

Societal implications. This work provides empirical evidence that the algorithmic foundation of the TSU—rejection-free block Gibbs sampling—achieves competitive or superior performance to standard MCMC baselines on our targeted problem classes. If the projected $10,000\times$ energy efficiency of dedicated thermodynamic hardware [7] is realised, the combination of algorithmic and hardware efficiency could substantially reduce the energy footprint of probabilistic inference and combinatorial optimisation workloads.

Future work. First, longer runs would establish whether THRML’s ESS/s advantage yields full convergence on grids. Second, direct energy measurement on TSU hardware would replace the ESS/s proxy and enable a concrete comparison with GPU energy costs. Third, extending the benchmark to other NP-hard classes—graph colouring, satisfiability, Bayesian inference—would show how broadly the block Gibbs advantage generalises.

APPENDIX A
SUPPLEMENTARY MATERIAL

This appendix collects the supplementary figures and tables referenced in Section IV. Figures 7–14 provide additional diagnostic, solution-quality, and scaling plots for the experiment groups in Section IV: MaxCut degree-3, the ferromagnetic and anti-ferromagnetic grids, and the ferromagnetic RRG. Table II reports the full convergence ($\hat{R}$), efficiency (ESS/s), and solution-quality (mean best) diagnostics for every configuration and sampler, pooled over the three runs (150 chains); and Table III gives the complete pairwise Mann-Whitney U comparisons, including the MH-vs-Gibbs comparison omitted from Table I, with exact two-sided p -values and rank-biserial effect sizes r .

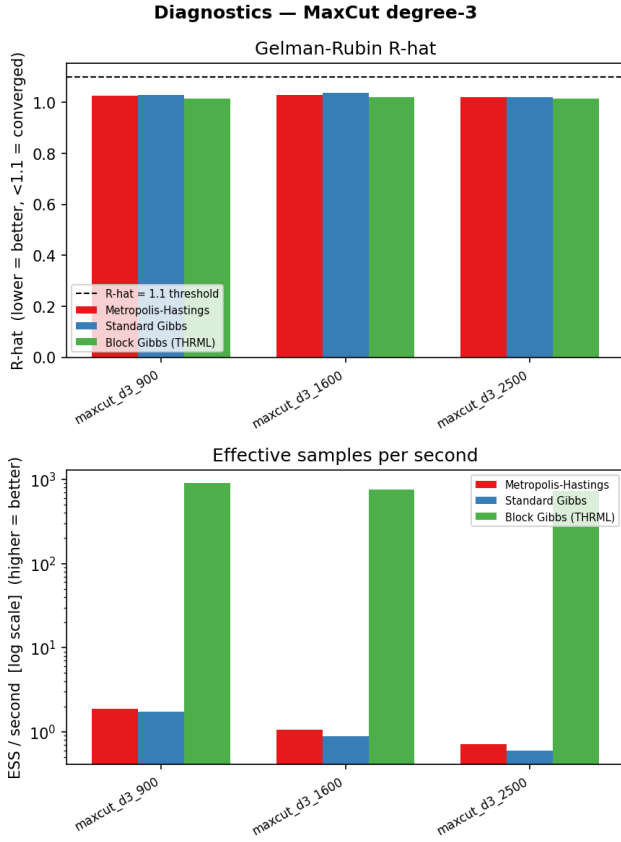


Fig. 7. Gelman–Rubin $\hat{R}$ (top) and ESS/s on log scale (bottom) for MaxCut degree-3 across system sizes.

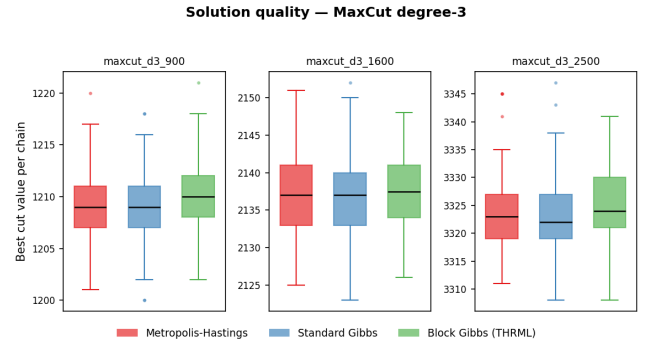


Fig. 8. Distribution of per-chain best cut values for MaxCut degree-3, one panel per system size with independent vertical axes. Box = interquartile range, line = median, whiskers = $1.5 \times \text{IQR}$, points = outliers. Higher is better.

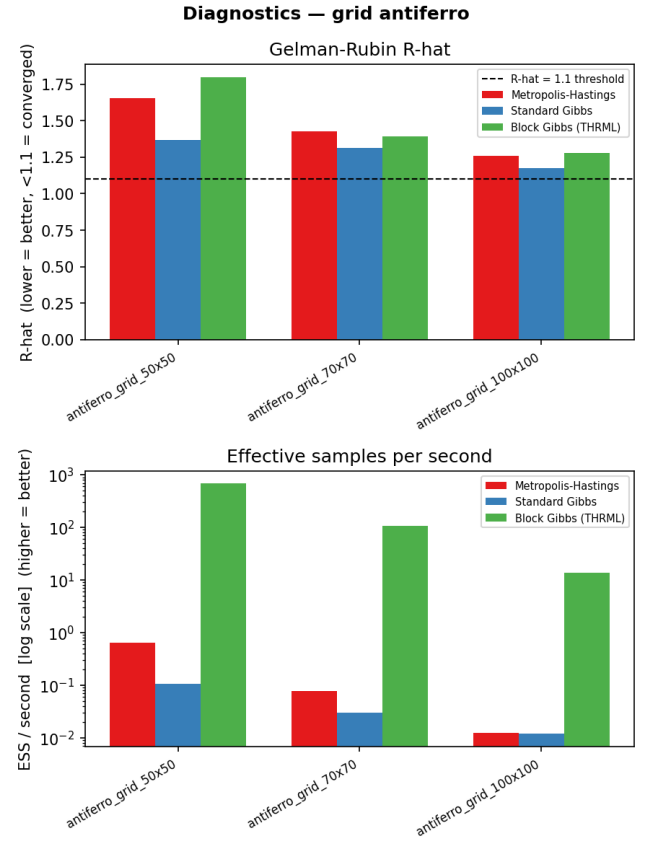


Fig. 9. Gelman–Rubin $\hat{R}$ (top) and ESS/s on log scale (bottom) for the anti-ferromagnetic grid across system sizes.

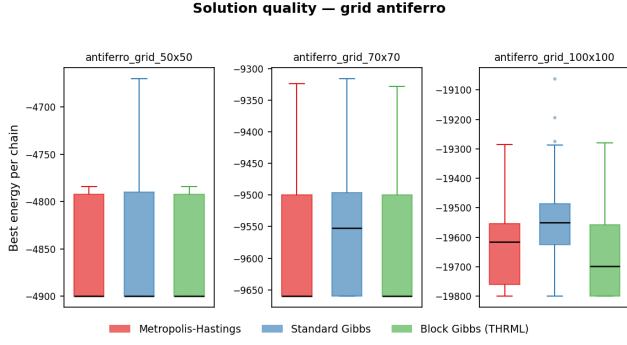


Fig. 10. Distribution of per-chain best energies for the anti-ferromagnetic grid, independent vertical axes per size. Lower is better.

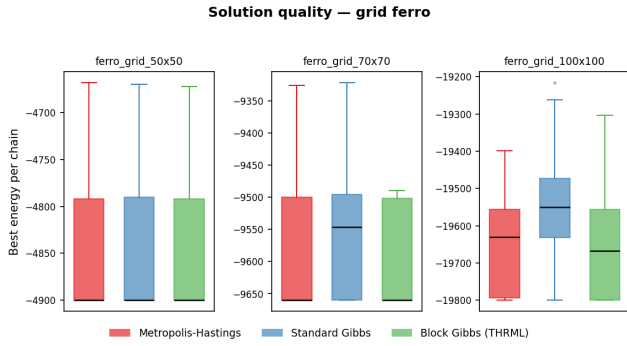


Fig. 11. Distribution of per-chain best energies for the ferromagnetic grid, independent vertical axes per size. Lower is better.

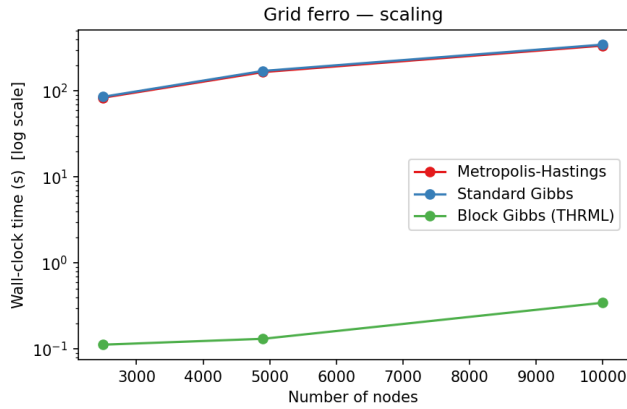


Fig. 12. Wall-clock time vs. number of nodes on log scale for the ferromagnetic grid. Each point is the mean across 3 runs.

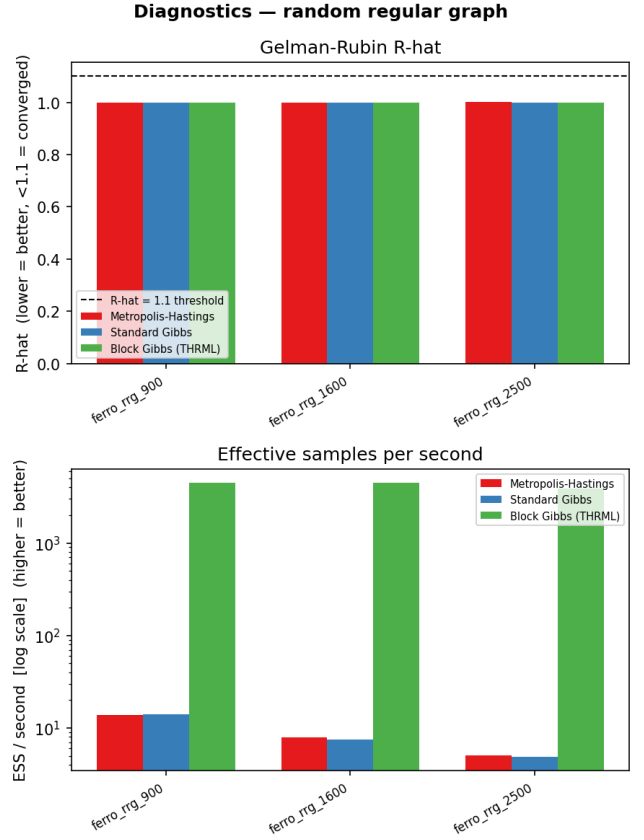


Fig. 13. Gelman–Rubin $\hat{R}$ (top) and ESS/s on log scale (bottom) for the ferromagnetic RRG across system sizes; $\hat{R} \leq 1.001$ confirms full convergence.

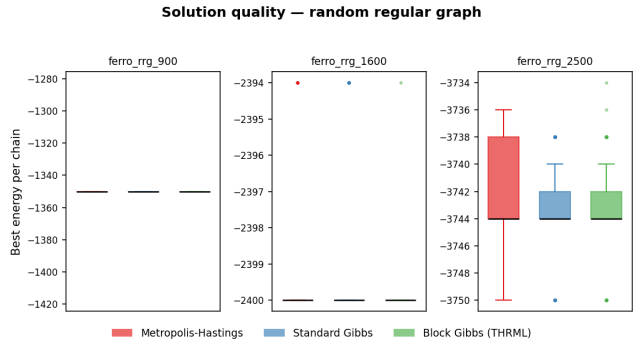


Fig. 14. Distribution of per-chain best energies for the ferromagnetic RRG. The near-complete overlap reflects the statistically equivalent solution quality reported in Section IV.

TABLE II

FULL CONVERGENCE AND EFFICIENCY DIAGNOSTICS FOR ALL 19 CONFIGURATIONS AND THREE SAMPLERS, POOLED OVER THREE RUNS (150 CHAINS). $\hat{R}$ IS THE GELMAN–RUBIN STATISTIC (< 1.1 INDICATES CONVERGENCE); ESS IS THE PER-CHAIN EFFECTIVE SAMPLE SIZE AND ESS/s THE EFFECTIVE SAMPLES PER SECOND; MEAN BEST IS THE MEAN PER-CHAIN BEST ENERGY (ISING, LOWER IS BETTER) OR CUT VALUE (MAXCUT, HIGHER IS BETTER). NOTE THAT ESS IS COMPARABLE ACROSS SAMPLERS WHILE ESS/s DIFFERS BY ORDERS OF MAGNITUDE.

Config	$\hat{R}$			ESS			ESS/s			Mean best		
	MH	Gibbs	THRML	MH	Gibbs	THRML	MH	Gibbs	THRML	MH	Gibbs	THRML
<i>Ferromagnetic grid</i> ($\beta = 1.0$)												
50×50	1.652	1.370	1.823	63.1	12.7	67.9	0.8	0.1	604.7	−4863.77	−4861.07	−4864.61
70×70	1.347	1.319	1.445	15.4	5.6	17.4	0.1	0.0	132.5	−9602.60	−9560.19	−9604.39
100×100	1.214	1.182	1.284	4.5	4.1	4.8	0.0	0.0	13.8	−19652.63	−19551.12	−19664.47
<i>Anti-ferromagnetic grid</i> ($\beta = 1.0$)												
50×50	1.656	1.369	1.797	53.8	9.1	73.2	0.6	0.1	683.3	−4865.51	−4859.65	−4869.00
70×70	1.428	1.311	1.391	12.5	4.9	14.2	0.1	0.0	108.6	−9596.49	−9564.01	−9595.89
100×100	1.259	1.176	1.280	4.1	4.1	4.6	0.0	0.0	13.7	−19630.15	−19557.77	−19680.24
<i>Temperature sweep</i> (70×70 ferromagnetic)												
$\beta = 0.1$	1.000	1.000	1.000	478.2	477.1	468.9	2.9	2.8	3462.2	−128.33	−128.53	−129.15
$\beta = 0.5$	1.155	1.130	1.164	24.1	15.4	34.4	0.1	0.1	261.6	−4238.97	−4212.35	−4244.63
$\beta = 1.0$	1.410	1.254	1.407	13.7	4.4	15.7	0.1	0.0	118.9	−9593.27	−9570.71	−9603.25
$\beta = 2.0$	1.359	1.291	1.522	6.9	4.5	7.1	0.0	0.0	53.8	−19230.03	−19156.93	−19186.72
<i>Ferromagnetic RRG</i> (degree 3, $\beta = 1.0$)												
900	1.000	1.000	1.000	427.9	423.0	466.0	13.9	14.1	4526.0	−1350.00	−1350.00	−1350.00
1600	1.000	1.000	1.000	432.5	416.9	469.8	7.9	7.5	4468.9	−2399.72	−2399.56	−2399.96
2500	1.001	1.000	1.000	423.8	426.2	474.0	5.0	4.9	3904.0	−3743.12	−3743.73	−3743.40
<i>MaxCut RRG, degree 3</i> ($\beta = 1.0$)												
900	1.028	1.030	1.016	54.2	52.9	85.8	1.9	1.7	897.8	1209.03	1208.97	1209.96
1600	1.031	1.037	1.020	57.1	50.1	79.6	1.1	0.9	755.5	2136.93	2136.74	2137.35
2500	1.022	1.022	1.015	61.0	51.9	88.6	0.7	0.6	726.4	3323.24	3322.77	3325.14
<i>MaxCut RRG, degree 5</i> ($\beta = 1.0$)												
900	1.055	1.060	1.049	61.6	56.0	91.2	1.9	1.8	671.3	1844.31	1844.66	1845.36
1600	1.088	1.088	1.070	33.2	31.9	55.9	0.6	0.6	385.7	3280.70	3281.05	3282.53
2500	1.063	1.064	1.078	23.9	18.2	38.3	0.3	0.2	228.7	5115.27	5114.41	5118.72

TABLE III

COMPLETE PAIRWISE MANN-WHITNEY U COMPARISONS ON THE 150 POOLED PER-CHAIN BEST VALUES. EACH PAIR OF COLUMNS GIVES THE TWO-SIDED p -VALUE AND THE RANK-BISERIAL EFFECT SIZE r . FOR ISING CONFIGURATIONS A NEGATIVE r FAVOURS THE FIRST-NAMED SAMPLER (LOWER ENERGY); FOR MAXCUT A POSITIVE r FAVOURS THE FIRST-NAMED SAMPLER (HIGHER CUT).

Config	THRML vs MH		THRML vs Gibbs		MH vs Gibbs	
	p	r	p	r	p	r
<i>Ferromagnetic grid</i>						
50×50	0.793	−0.014	0.300	−0.058	0.421	−0.045
70×70	0.781	−0.016	< 0.001	−0.312	< 0.001	−0.297
100×100	0.252	−0.076	< 0.001	−0.480	< 0.001	−0.450
<i>Anti-ferromagnetic grid</i>						
50×50	0.588	−0.029	0.040	−0.114	0.117	−0.088
70×70	0.965	−0.003	< 0.001	−0.220	< 0.001	−0.226
100×100	0.001	−0.214	< 0.001	−0.515	< 0.001	−0.334
<i>Temperature sweep</i> (70×70 ferromagnetic)						
$\beta = 0.1$	0.065	−0.123	0.231	−0.080	0.535	0.042
$\beta = 0.5$	0.337	−0.064	< 0.001	−0.308	< 0.001	−0.252
$\beta = 1.0$	0.294	−0.062	< 0.001	−0.253	0.003	−0.189
$\beta = 2.0$	0.028	0.125	0.002	−0.190	< 0.001	−0.323
<i>Ferromagnetic RRG</i> (degree 3)						
900	1.000	0.000	1.000	0.000	1.000	0.000
1600	0.032	−0.040	0.003	−0.067	0.332	−0.027
2500	0.484	−0.043	0.390	0.052	0.127	0.093
<i>MaxCut RRG, degree 3</i>						
900	0.027	0.147	0.020	0.155	0.910	0.008
1600	0.409	0.055	0.174	0.091	0.658	0.030
2500	0.003	0.199	< 0.001	0.222	0.750	0.021
<i>MaxCut RRG, degree 5</i>						
900	0.013	0.165	0.157	0.094	0.330	−0.065
1600	0.012	0.168	0.038	0.138	0.670	−0.028
2500	< 0.001	0.265	< 0.001	0.336	0.265	0.074

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